



AGRICOOOL - By Agrinova

A farmers friend -
किसान मित्र

A electricity free solution for
your ताजा उपज

Presented by

**Sana and
Khwaish**



Our vision

Why we made AgriCool

AgriCool is an affordable, sustainable cold storage solution designed to reduce post-harvest losses for farmers in Gujarat. By using evaporative cooling technology, AgriCool maintains the freshness of perishable produce without electricity. This low-cost alternative to traditional cold storage is for farmers facing high energy costs and spoilage losses.

AgriCool is priced at **₹20,000 per unit with a capacity of 1500– 1600 kgs** of produce and has a volume of **2.16m³ (1.5m x 1.2m x 1.2m)**, making it cheaper than traditional cold storage, and unlike cold storage, it is a **one-time purchase** and does not require annual payments. It requires **no electricity** and functions by utilizing a water-soaked fabric covering, which **cools stored produce through evaporation**. This ensures efficient storage, especially in arid climates like Gujarat.

· **Post-Harvest Losses:** According to a PIB (Post Information Bureau) article published on December 2022 on post-harvest losses, India loses 6.02%–15.05% of fruits and 4.87%–11.61% of vegetables due to inadequate storage, leading to an annual loss of billions.

Source: https://pib.gov.in/PressReleaselframePage.aspx?PRID=1885038&utm_source=google.com

High Cost of Cold Storage: Based on a source from Maharashtra State Warehousing, traditional cold storage costs ₹2,500 per ton per month for frozen items, when calculated at ₹30000 per ton annually, making it inaccessible to small farmers and expensive in the long run.

Source: https://mswarehousing.com/Mar-mswc/wp-content/uploads/2021/04/Panvel_Cold_storage- schedule.pdf?utm_source=google.com

· **Energy Dependency:** Cold storage units are concentrated in a few states and require constant electricity, which is unreliable in rural areas and harmful to the planet.



The solution

Studio
ShodWe

Product Overview

AgriCool is a passive evaporative cooling storage system that maintains temperatures lower than ambient temperatures (**about 7-15 Degrees Celsius**), extending shelf life and reducing waste.

How It Works

- The unit is made of a **breathable shell** protecting the **internal cooling** system and is wrapped around a metal frame.
- Water is poured into the second layer with a **hydrogel bead sheet** that **swells** when soaked and **releases moisture**.
- As water evaporates, it cools the internal chamber, reducing the temperature.
- Farmers need to **re-soak** the unit every **8-12 hours**, depending on humidity levels.



Why it works

- Produce doesn't get spoiled, as the hydrogel beads create a cooling effect that **slows down microbial activity**.
- Cotton **prevents condensation buildup** and does not allow bacteria or mould to form.
- Hydrogel beads absorb up to three times their water content; in this case, nearly **7500l** and reduce temperatures by **7-15 Degrees Celsius**
- Hydrogel beads **release moisture as vapour** rather than dripping water, ensuring it won't seep through



STRUCTURE

4 layers of sustainability



Outer Protective Layer – The Structural Shell

Material: Breathable Jute Function:

- Acts as a **protective casing**, shielding the inner cooling system from external dirt, dust, and mechanical damage.
- Highly porous, allowing **air circulation** to facilitate evaporation.
- Durable to withstand repeated use in rough agricultural conditions.
- While jute is porous to allow air circulation for efficient evaporation, its fibrous it promotes the **evaporation process on the surface**, ensuring that water does not simply seep inward
 - It is **naturally reflective** due to the presence of cellulose and lignin,



Moisture-Retaining Layer – The Cooling System

Material: Absorbent sheet layer (Hydrogel beads)

Function:

- Stores and retains water** to sustain the evaporative cooling process.
- As water evaporates from this layer, it removes **latent heat**, creating a temperature drop in the inner chamber as it draws heat energy from its surroundings.
- This gel structure **traps the water molecules** within a cross-linked polymer network, releasing water slowly as vapor rather than as free-flowing liquid.



Insulating Layer – Temperature Stabilization

Material: Coconut husk fibers

Function:

- Prevents heat gain** from the external environment.
- Helps retain the cooling effect produced by the evaporative layer.
- Ensures a stable temperature** for perishable goods.
- Lightweight, reducing the overall weight of the AgriCool unit



Inner Storage Chamber – The Produce Store

Material: Food-grade, breathable fabric (organic cotton)

Function:

- Provides a **clean and hygienic space** for storing fresh produce.
- Maintains humidity levels to prevent dehydration** of stored crops.
- Ensures maximum airflow to enhance cooling efficiency.
- Prevents direct moisture** contact with produce



Advantages



No electricity needed

Cost effective

Sustainable

Lightweight

•No Electricity Needed

- Works without power, making it **ideal for rural areas and developing regions**.
- Farmers don't need to invest in expensive refrigeration units.

•Cost-Effective & Scalable

- Cheaper than conventional cold storage systems, **only requiring a one-time payment**.
- It can be mass-produced using locally available materials.

•Eco-Friendly & Sustainable

- Made from **biodegradable materials** like jute, hemp, and cotton.
- Uses only water** to function, reducing the carbon footprint.
- Saves on waste

•Lightweight & Portable

- It can be **easily carried and transported** to markets or homes.

Farmers who use cold-storage

- If produce is stored with high field heat due to its high respiration rate, the cooling system uses more energy which leads to uneven cooling, condensation, and faster spoilage.
- AgriCool Sack removes field heat before cold storage, lowering the temperature naturally.
- Cold storage is most effective when crops are pre-cooled before being placed inside.

•Source:

<https://openknowledge.fao.org/handle/20.500.14283/ca6640en>

How we expand

Summary



Unique Value Proposition (UVP)

- Cost-effective: It is cheaper than conventional electric cold storage.
- Energy-independent: Uses evaporative cooling; requires only water.
- Scalable & Easy to Use: Requires no technical expertise for operation.

Supply Chain & Production:

- Marketing: Live demonstrations and Social media and digital outreach
- Manufacturing: Set up in Ahmedabad using small-scale production units.
- Distribution: Partnerships with local agricultural cooperatives and Agri-tech startups.



Key Objectives

Target Market:

- **Medium to large-scale** farmers in Gujarat.
- **Local vegetable vendors** and cooperatives.
- Government and NGO initiatives promoting sustainable farming.

Market Demand & Trends:

- Growing awareness about food loss reduction.
- Growing awareness around climate change.
- Increased adaptation of low-cost storage solutions in developing regions.

Farmer feedback

Farmers Feedback:

- 10 farmers from Gujarat were asked for their feedback
 - 8/10 were satisfied with the cost of the product
- 6/10 were willing to buy the product after successful and permitted testing
 - Farmers with over 20 acres of land said it would be beneficial for them to maintain export-level quality

Costings



Component	Material	Final cost (₹)
Steel frame Source: https://www.padmawatitraders.in/	Mild steel pipes (6 kg total, 2–3 cm diameter, 14.1m) – provide structural support and maintain shape Charged at: ₹52/- per kg for Mass required per unit: 14.1m / 6m = 2.35 Total cost: 2.35 x ₹52 = ₹156	₹156
Outer Layer Source: https://www.sumatikhadifabric.com/cotton-jute-fabric.html	Jute fabric (17.2 m ²) – (400gsm) Breathable and reflective, protects inner layers from direct sunlight. Charged at: ₹34/- per meter LSA of unit = 2x1.5(1.2 + 1.2) = 17.2m ² Total cost: 17.2 x ₹34 = ₹244.8	₹244.8
Middle Layer Source: https://www.indiamart.com/prod-detail/super-absorbent-polymer-for-ice-gel-packs-6266555888.html	Hydrogel beads (super absorbent polymer) (25 kg) – Retains water for sustained evaporative cooling Charged at: ₹170/ kg Total cost: 25 x ₹170 = 4250	₹4250
Insulation layer Source: https://www.indiamart.com/prod-detail/coconut-coir-fibre-20546868848.html?vcdingform=1	Coconut fiber padding (0.5 cm thick) – Acts as a thermal barrier to reduce external heat transfer Charged at: ₹20/ kg Volume needed: 17.2m ² x 0.005m = 0.344m ³ Weight = 0.344m ³ x 835kg/m ³ = 287.24kg Total cost: 287.24kg x ₹20	₹5744.8
Inner layer Source: https://shorturl.at/1xvuP	Cotton fabric (1.54kg, organic) – Provides a hygienic, breathable lining for the storage chamber Charged at: ₹130/ kg LSA of inner chamber = 2 x 1.4m(1.1m+1.1m) = 6.16m ² Mass needed: 6.16m ² x 0.25kg/m ² = 1.54kg	₹200.2

Costings



	Total cost: $1.54\text{kg} \times ₹130 = ₹200.2$	
Seal flap & fastening Source: https://www.sumatikhadifabric.com/cotton-jute-fabric.html https://shorturl.at/1xvuP	Dual-layered flap (inner cotton + outer insulated fabric with a waterproof sealing strip) secured with Velcro Total Costing: -Upper jute : $1.44\text{m}^2 \times 0.4\text{kg}/\text{m}^2 = 0.576\text{ kg}$ - $0.576\text{kg} \times ₹52 = ₹30$ -Inner cotton: $1.44\text{m}^2 \times 0.24\text{kg}/\text{m}^2 = 0.346\text{kg}$ - $0.346\text{kg} \times ₹130 = ₹45$ -Velcro: ₹120/ meter - $1.1 \times 1.1 = 1.21\text{m}^2$ - $1.21 \times ₹120 = ₹145.2$	₹220.2
Labor & Assembly	Skilled stitching, layering, and integration of the steel frame and all fabric layers	₹2500
Miscellaneous costs	Packaging, transport, and marketing	₹2000
Total cost per unit	-	₹15,316

Dimensions:

- $1.5\text{m} \times 1.2\text{m} \times 1.2\text{m}$ (Outer layer)
- $1.4\text{m} \times 1.1\text{m} \times 1.1\text{m}$ (Inner storage)
- $1.5\text{m} \times 1.2\text{m} \times 1\text{cm}$ (Flap dimensions)

Selling Price:

- 30% profit margin for sustainability
- Selling Price ₹20,000
- Affordable for farmers while being profitable



Thank You

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काम

